

Certifying the decidability of the word problem in monoids, at large

Reinis Cirpons

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A data driven approach to mathematics

1. Solve some small examples (by hand or computationally),
2. spot a common pattern,
3. generalize,
4. prove the generalization works.

n	$ \mathcal{T}_n $	max. right	max. left
1	1	0	0
2	4	3	1
3	27	7	4
4	256	15	14
5	3125	≥ 31	≥ 51
n	n^n	$2^n - 1?$	$B(n) - 1?$

Need for certification

Bad data → wrong insights → no proof or false proof

Need for certification

Bad data $\rightarrow$ wrong insights $\rightarrow$ no proof or false proof

Question

How can we certify the correctness of computational solutions of instances of a mathematical problem when the computational tools are not themselves formally verified? Can this be done efficiently and at scale?

The Word Problem in Monoids

Disclaimer

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SURVEY



The word problem for one-relation monoids: a survey

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Abstract

This survey is intended to provide an overview of one of the oldest and most celebrated open problems in combinatorial algebra: the word problem for one-relation monoids.

The word problem

$$a, b \in M$$

$$b a a b a = a$$

$$b b a a b a a b a \stackrel{?}{=} a$$

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$$\boxed{b} \boxed{a} \boxed{a} \boxed{b} \boxed{a} = \boxed{a}$$

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$$\boxed{a} \overset{?}{=} \boxed{a}$$

Word problem

Presentation: $\langle \underbrace{a, b}_{\text{generators}} \mid \underbrace{baaba = a}_{\text{relations}} \rangle$.

Definition (Word problem for $\langle A \mid R \rangle$)

Given: words

$$w_1, w_2 \in A^*.$$

Decide whether:

$$w_1 = w_2 \text{ modulo } R.$$

Undecidability

Is the word problem decidable for **every** presentation $\langle A \mid R \rangle$ of:

- ▶ Monoids (associative law and unit)?

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- ▶ Groups (associative law, unit and inverses)?

Undecidability

Is the word problem decidable for **every** presentation $\langle A \mid R \rangle$ of:

- ▶ Monoids (associative law and unit)? **No.**
[Markov 1947] and [Post 1947]
- ▶ Groups (associative law, unit and inverses)? **No.**
[Novikov 1955], [Boone 1958] and [Britton 1958]

Small examples

Theorem (Tseytin 1958)

The presentation $\langle a, b, c, d, e \mid R \rangle$ with 5 generators and 7 relations:

$$R = \{ac = ca, \quad ad = da, \quad bc = cb, \quad bd = db, \\ eca = ce, \quad edb = de, \quad cca = ccae\}$$

defines a monoid with undecidable word problem.

Small examples

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The smallest undecidable examples in terms of the number of relations for

- ▶ Monoids: 2 generators and **3** relations. [Matiyasevich 1967]
- ▶ Groups: 5 generators and **12** relations. [Borisov 1969]

1-relation presentations

Is the word problem decidable for every presentation $\langle A \mid \mathbf{u} = \mathbf{v} \rangle$ of:

- ▶ Groups (associative law, unit and inverses)?
- ▶ Monoids (associative law and unit)?

1-relation presentations

Is the word problem decidable for every presentation $\langle A \mid \mathbf{u} = \mathbf{v} \rangle$ of:

- ▶ Groups (associative law, unit and inverses)? **Yes.**

[Magnus 1932]

- ▶ Monoids (associative law and unit)?

1-relation presentations

Is the word problem decidable for every presentation $\langle A \mid \mathbf{u} = \mathbf{v} \rangle$ of:

- ▶ Groups (associative law, unit and inverses)? **Yes.**

[Magnus 1932]

- ▶ Monoids (associative law and unit)? **Open.**

Computational enumeration

Mission

Question

What is the “smallest” example of a 1-relation monoid where decidability of the word problem is unknown?

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What is the “smallest” example of a 1-relation monoid where decidability of the word problem is unknown?

Goal: Solve all

$$\langle a, b \mid u = v \rangle \text{ with } |u|, |v| \leq 10.$$

Total of $(2^{11} - 1)^2 = 4.2 \cdot 10^6$ presentations.

Prior art: Pedersen 1989 for $|u|, |v| \leq 6$.

Symmetries

Decidability of the word problem is unchanged by:

1. swapping relation words,

$$\langle a, b \mid u = v \rangle \rightsquigarrow \langle a, b \mid v = u \rangle$$

2. permuting the alphabet, $a \mapsto b$ and $b \mapsto a$, e.g.

$$\langle a, b \mid baabaa = aabab \rangle \rightsquigarrow \langle a, b \mid abbabb = bbaba \rangle$$

3. reversing relation words, e.g.

$$\langle a, b \mid baabaa = aabab \rangle \rightsquigarrow \langle a, b \mid aabaab = babaa \rangle$$

$$4.2 \cdot 10^6 \rightarrow 5.2 \cdot 10^5$$

Reductions

Theorem (Adian and Oganesian 1987)

For all sets A and words $u, v \in A^$ there exists a pair of words $u', v' \in \{a, b\}^*$ such that the word problem of $\langle A \mid u = v \rangle$ reduces to the word problem of a presentation of one of the following forms*

$$\langle a, b \mid bu'a = av'a \rangle \quad \text{or} \quad \langle a, b \mid bu'a = a \rangle$$

for some $u', v' \in \{a, b\}^$.*

$$5.2 \cdot 10^5 \rightarrow 2.6 \cdot 10^5$$

Decidability criteria

1. a presentation $\langle a, b \mid u = v \rangle$ is *special* [35, Section 2.3], i.e., $v = \varepsilon$;
2. a presentation $\langle a, b \mid u = v \rangle$ is *length preserving*, i.e., $|u| = |v|$;
3. a presentation $\langle a, b \mid u = v \rangle$ is *cycle-free* [35, Section 2.2], i.e., the initial (respectively terminal) letters of u and v differ;
4. a presentation is the *free product* of monogenic monoids, i.e., both sides of each relation are powers of the same generator;
5. a presentation satisfies the *first Watier condition* [35, Theorem 4.10];
6. a presentation is left-cycle free and *stabilizes a generator* [35, Theorem 4.9], i.e., the initial letters of u and v differ and there is a letter which occurs an equal number of times in u and v ;
7. a presentation satisfies the $C(3)$ small overlap criterion [39], see also [32] for a description of the linear-time procedure implemented in libsemigroups;
9. the Knuth-Bendix algorithm terminates with respect to some lenlex ordering, with a timeout of 10 milliseconds per Knuth-Bendix instance;
10. after recursively replacing a subword with a generator (using the T3 Tietze transformation) up to depth 5, the Knuth-Bendix algorithm terminates with respect to some lenlex ordering, with a timeout of 10 milliseconds per Knuth-Bendix instance;
11. a backtracking version of Knuth-Bendix, similar to [44, 45], terminates, with a timeout of 20 minutes per instance;
12. a rewriting system derived from Adian's algorithm [10] terminates according to the termination provers matchbox [43] and MultumNonMultum [24].

Computational results

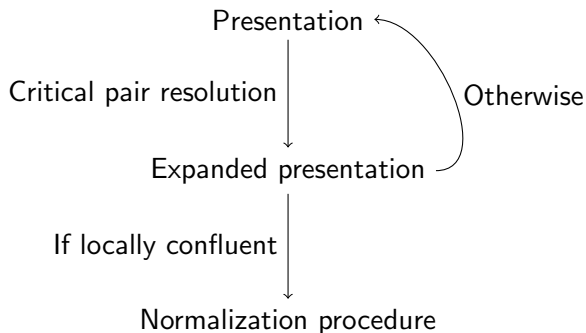
Method	Count
Known mathematical results	222786
Knuth-Bendix	37624
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Unsolved	260
Total	261632

Computational results

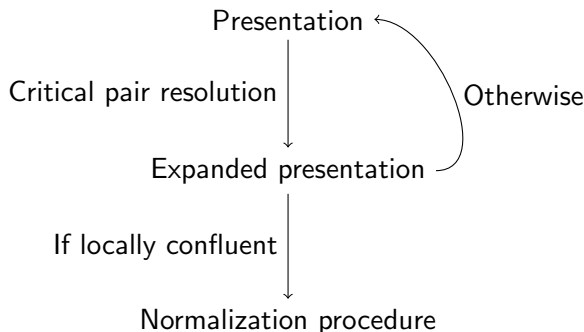
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Total time taken: 6 days and 14 hours, 6 days 7 hours spent on Knuth-Bendix based criteria.

The Knuth-Bendix algorithm



The Knuth-Bendix algorithm



Many applications in combinatorial semigroup theory: word problem, finiteness, low-dimensional homotopy, etc. Implemented in various computational algebra software: GAP, MAGMA, KBMag, libsemigroups etc.

Formalization

Formalizing an implementation of Knuth-Bendix is hard (the one found in `libsemigroups` is over 1600 lines of C++).

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Solution: Get a trace of the computation performed by Knuth-Bendix (steps taken to expand the presentation) and turn it into a formal certificate of correctness in Rocq!

$$\langle a, b \mid babaabaa = abaaabaaa \rangle$$

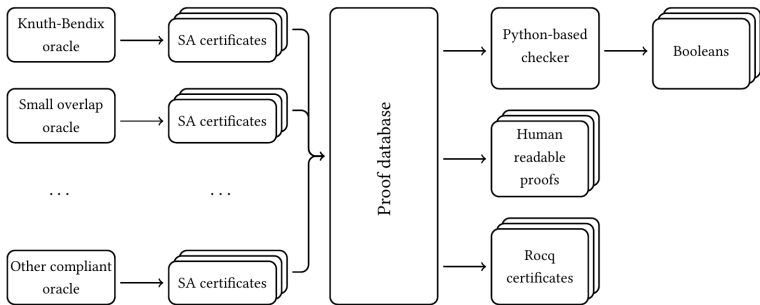
Definition present_entry := make_pres [::0;1]
 [:: ([::1;0;1;0;0;1;0;0], [::0;1;0;0;0;1;0;0;0]).

Knuth-Bendix + Heuristics

$$\left\langle a, b, c \mid \begin{array}{l} baaa = c, \\ babaac = acca, \\ babaabaa = acc \end{array} \right\rangle$$

Definition present_final := make_pres [::0;1;2]
 [:: ([::1;0;0;0], [::2]);
 ([::1;0;1;0;0;2], [::0;2;2;0]);
 ([::1;0;1;0;0;1;0;0], [::0;2;2]).

Definition cert : pres_cert := [:: add_gen 2 [::1;0;0;0];
 add_rel [::1;0;1;0;0;2] [::0;2;2;0]
 [:: RTriple 1 5 false;
 RTriple 0 0 true;
 RTriple 1 1 true;
 RTriple 1 2 true];
 add_rel [::1;0;1;0;0;1;0;0] [::0;2;2]
 [:: RTriple 0 0 true;
 RTriple 1 1 true;
 RTriple 1 2 true];
 rm_rel 0
 [:: RTriple 2 0 true; I
 RTriple 0 1 false;
 RTriple 0 5 false]].



Need for speed

Certificates can get very large (1000s of lines)! Need to consider efficiency of the formalization:

- Tries for rewriting:

Naive: quadratic time → Tries: linear time

- `native_compute` machinery for normalization of terms:

Presentation	Baseline	<code>vm_compute</code>	<code>native_compute</code>
Renner D_4	7.7s	0.6s	0.3s
Renner D_5	132.0s	9.4s	2.4s

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With these optimizations, checking the correctness of all of the generated Rocq certificates takes about 7 hours on a single core (30 minutes with parallelism). Of this about 2.5 hours is spent parsing the presentation definitions and about 4.5 hours are spent checking the proof certificates.

Formalization results

Type	number	certified
AlphabetIsom	484925	103882
AlphabetReorder	168226	827
CompleteRewritingSystem	55330	55330
CycleFree	70547	0
EqualNumberOfOccurrences	56465	0
FlipAllRelations	385762	39813
FreeProductMonogenicAndFree	45	45
Monogenic	0	0
Reverse	259065	633
SmallOverlap	31204	0
Special	0	0
StronglyCompressAndReduce	502843	0
StronglyCompressToSpecial	555	0
Watier	44394	0
Total	2059361	200530

Rocq formalization of the theory (so far), computational trace database and trace to Rocq proof conversion script are available at:

`https://github.com/hivert/MonoidPresentation`

Library is useful more broadly, e.g. for proving correctness of presentations for monoids from the literature.

Future work

- ▶ Formalize more of the underlying mathematical theory of the word problem for 1-relation monoids;
- ▶ Improve efficiency of the Rocq formalization even more (native arrays for tries, other datastructures);
- ▶ Release the database in the form of an online Encyclopedia;
- ▶ A Polymath-style collaborative project for the word-problem in 1-relation monoids?

```

Variant prescertificate :=
  (* requires: rewriting certificate + termination ordering *)
  | CompleteRewritingSystem of pres_cert & seq Alph
  (* requires: the repeated letter a in  $\langle a \ b \mid a^k = a^l \rangle$  *)
  | EqualNumberOfOccurrences of Alph
  | Monogenic
  [ ... ten more cases ... ]
  (* Recursive certificates: reduction proofs to another presentation *)
  (* apply rev to all relation words keeping the gens and relation order *)
  | Reverse of recursive_certificate
  (* requires: the word which is retained and the letter it is sent to *)
  | StronglyCompressAndReduce of recursive_certificate &
    word & Alph.

```

References I

-  Adian, S.I. and Oganesian, G.U. (1987). “On the word and divisibility problems for semigroups with one relation”. In: *Math. Notes* 41. [Mat. Zametki 41, 3 (1987)], pp. 235–240.
-  Boone, William W. (Oct. 1958). “The Word Problem”. In: *Proceedings of the National Academy of Sciences* 44.10, pp. 1061–1065. DOI: 10.1073/pnas.44.10.1061.
-  Borisov, V. V. (Nov. 1969). “Simple examples of groups with unsolvable word problem”. In: *Mathematical Notes of the Academy of Sciences of the USSR* 6.5, pp. 768–775. ISSN: 1573-8876. DOI: 10.1007/bf01101402.
-  Britton, J. L. (1958). “The Word Problem for Groups”. In: *Proceedings of the London Mathematical Society*. 3rd ser. 8.4, pp. 493–506. DOI: 10.1112/plms/s3-8.4.493.
-  Magnus, W. (Dec. 1932). “Das Identitätsproblem für Gruppen mit einer definierenden Relation”. In: *Mathematische Annalen* 106.1, pp. 295–307. ISSN: 1432-1807. DOI: 10.1007/bf01455888.

References II

-  Markov, A. A. (1947). “On the impossibility of certain algorithms in the theory of associative systems”. In: *Doklady Akad. Nauk SSSR (N.S.)* 55, pp. 583–586.
-  Matiyasevich, Yuri V. (1967). “Simple examples of undecidable associative calculi”. English. In: *Soviet Mathematics. Doklady* 8, pp. 555–557. ISSN: 0197-6788.
-  Novikov, P. S. (1955). “On the algorithmic unsolvability of the word problem in group theory”. In: *Trudy Mat. Inst. Steklov.* 44, pp. 3–143. URL: <http://mi.mathnet.ru/tm1180>.
-  Nyberg-Brodda, Carl-Fredrik (Aug. 2021). “The word problem for one-relation monoids: a survey”. In: *Semigroup Forum* 103.2, pp. 297–355. DOI: 10.1007/s00233-021-10216-8.
-  Pedersen, John (1989). “Morphocompletion for one-relation monoids”. In: *Rewriting Techniques and Applications*. Ed. by Nachum Dershowitz. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 574–578. DOI: 10.1007/3-540-51081-8_141.

References III

-  Post, Emil L. (1947). "Recursive unsolvability of a problem of Thue". In: *J. Symbolic Logic* 12, pp. 1–11. ISSN: 0022-4812. DOI: 10.2307/2267170. URL: <https://doi-org.uea.idm.oclc.org/10.2307/2267170>.
-  Tseytin, Gregory S. (1958). "An associative calculus with an insoluble equivalence problem". In: *Trudy Mat. Inst. Steklov.* 52.